

Paste Key:

$\Sigma, \ddot{a}, \mathfrak{b}, \dot{c}, \in, \equiv, \oplus$

HW5: Number Theory and RSA

CYB-402

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1.  $n = 88$ , symmetric cipher so same key is used by both parties.

,every person will generate their own key pair once so no need for a new key per communication air

$\frac{n}{2}$  binomial coefficient of unordered pairs

$$\frac{n}{2} = n = \frac{n(n-1)}{2}$$

$$\frac{88}{2} = \frac{88(87)}{2} = \frac{7656}{2} = 3828 \text{ keys needed}$$

2.  $n = 88$ , RSA so asymmetric cipher each user has their own key pair

,every person will generate their own key pair once and can use a person's public key to send encrypted messages to them.

,no need for a new key per communication pair

Each employee will need 1 RSA key pair

So, total number of key pairs needed is 88.

3.

3.1.  $\gcd(3987, 2683)$

$$3987 = 1 * 2683 + 1304$$

$$2683 = 2 * 1304 + 75$$

$$1304 = 17 * 75 + 29$$

$$75 = 2 * 29 + 17$$

$$29 = 1 * 17 + 12$$

$$17 = 1 * 12 + 5$$

$$12 = 2 * 5 + 2$$

$$5 = 2 * 2 + 1$$

$$2 = 2 * 1 + 0$$

$$\gcd(3987, 2683) = \gcd(2683, 1304) = \gcd(1304, 75) = \gcd(75, 29) = \gcd(29, 17) = \gcd(17, 12) = \gcd(12, 5) = \gcd(5, 2) = \gcd(2, 1) = 1$$

$$\gcd(3987, 2683) = 1$$

$$3.2. \gcd(6393, 1234)$$

$$6393 = 5 * 1234 + 223$$

$$1234 = 5 * 223 + 119$$

$$223 = 1 * 119 + 104$$

$$119 = 1 * 104 + 15$$

$$104 = 6 * 15 + 14$$

$$15 = 1 * 14 + 1$$

$$14 = 14 * 1 + 0$$

$$\gcd(6393, 1234) = \gcd(1234, 223) = \gcd(223, 119) = \gcd(119, 104) = \gcd(104, 15) = \gcd(15, 14) = \gcd(14, 1) = 1.$$

$$\gcd(6393, 1234) = 1$$

4. initialize  $r_0, r_1$

$$S_0 = 1, t_0 = 0 \text{ so } r_0 = 1 * r_0 + 0 * r_1$$

$$S_1 = 0, t_1 = 1 \text{ so } r_1 = 0 * r_0 + 1 * r_1$$

$$r_{i-1} = q_i r_i + r_{i+1}, \text{ update } s_{i+1} = s_{i-1} - q_i s_i, t_{i+1} = t_{i-1} - q_i t_i$$

4.1 568 and 397

| i | $r_{i-1}$ | $r_i$ | $q_i$ | $s_i/t_i$  |
|---|-----------|-------|-------|------------|
| 1 | 568       | 397   |       | (0,1)      |
| 2 | 397       | 171   | 1     | (1,-1)     |
| 3 | 171       | 55    | 2     | (-2,3)     |
| 4 | 55        | 6     | 3     | (7,-10)    |
| 5 | 6         | 1     | 9     | (-65,93)   |
| 6 | 1         | 0     | 6     | (397,-568) |

**Computation and Check**

Division  $r_0 = 568, r_1 = 397$

$$568 = 1 * 397 + 171$$

$$397 = 2 * 171 + 55$$

$$171 = 3 * 55 + 6$$

$$55 = 9 * 6 + 1$$

$$6 = 6 * 1 + 0$$

$s_i/t_i$  steps

$$s_0 = 1, t_0 = 0, s_1 = 0, t_1 = 1$$

$$i = 1: s_2 = s_0 - q_1 s_1 = 1 - 1 * 0 = 1; t_2 = t_0 - q_1 t_1 = 0 - 1 * 1 = -1$$

$$i = 2: s_3 = s_1 - q_2 s_2 = 0 - 2 * 1 = -2; t_3 = t_1 - q_2 t_2 = 1 - 2 * -1 = 3$$

$$i = 3: s_4 = s_2 - q_3 s_3 = 1 - 3 * -2 = 7; t_4 = t_2 - q_3 t_3 = -1 - 3 * 3 = -10$$

$$i = 3: s_5 = s_3 - q_4 s_4 = -2 - 9 * 7 = -65; t_5 = t_3 - q_4 t_4 = 3 - 9 * -10 = 93$$

$i = 5$ : remainder zero so ignore

$$s = -65, t = 93$$

Check

$$s * r_0 + t * r_1 = \gcd(r_0, r_1)$$

$$(-65) * 568 + 93 * 397 = 36,920 - 36,921 = 1$$

#### 4.2. 3793 and 1753

| i | $r_{i-1}$ | $r_i$ | $q_i$ | $s_i/t_i$    |
|---|-----------|-------|-------|--------------|
| 1 | 3793      | 1753  |       | (0,1)        |
| 2 | 1753      | 287   | 2     | (1,-2)       |
| 3 | 287       | 31    | 6     | (-6,13)      |
| 4 | 31        | 8     | 9     | (55,-119)    |
| 5 | 8         | 7     | 3     | (-171,370)   |
| 6 | 7         | 1     | 1     | (226,-489)   |
| 7 | 1         | 0     | 7     | (-1753,3793) |

#### Computation and Check

Division  $r_0 = 3793, r_1 = 1753$

$$3793 = 2 * 1753 + 287$$

$$287 = 9 * 31 + 8$$

$$31 = 3 * 8 + 7$$

$$8 = 1 * 7 + 1$$

$$7 = 7 * 1 + 0$$

$s_i/t_i$  steps

$$s_0 = 1, t_0 = 0, s_1 = 0, t_1 = 1$$

$$i = 1: s_2 = 1 - 2 * 0 = 1; t_2 = 0 - 2 * 1 = -2$$

$$i = 2: s_3 = 0 - 6 * 1 = -6; t_3 = 1 - 6 * -2 = 13$$

$$i = 3: s_4 = 1 - 9 * -6 = 55; t_4 = -2 - 9 * 13 = -119$$

$$i = 4: s_5 = -6 - 3 * 55 = -171; t_5 = 13 - 3 * -119 = 370$$

$$i = 5: s_6 = 55 - 1 * -171 = 226; t_6 = 13 - 3 * -119 = -489$$

$i = 6$ : remainder zero so ignore

$$s = 226, t = -489$$

Check

$$s * r_0 + t * r_1 = \gcd(r_0, r_1)$$

$$226 * 3793 + (-489) * 1753 = 857,218 - 587,217 = 1$$

5

$$5.1 \ a = 6, m = 26$$

| <b>i</b> | <b><math>r_{i-1}</math></b> | <b><math>r_i</math></b> | <b><math>q_i</math></b> | <b><math>s_i/t_i</math></b> |
|----------|-----------------------------|-------------------------|-------------------------|-----------------------------|
| <b>0</b> |                             | 26                      |                         | (1,0)                       |
| <b>1</b> | 26                          | 6                       | 4                       | (0,1)                       |
| <b>2</b> | 6                           | 2                       | 3                       | (1,-4)                      |
| <b>3</b> | 2                           | 0                       |                         | (-3,13)                     |

Computation

$$26 = 4 * 6 + 2$$

$$6 = 3 * 2 + 0$$

Since the last nonzero remainder  $r_2 = 2$  and  $\gcd(26,6) = 2$ , no inverse of 6 mod 26 exists.

Check

$$1 * 26 + (-4) * 6 = 2$$

5.2  $a = 18, m = 1000$

| i | $r_{i-1}$ | $r_i$ | $q_i$ | $s_i/t_i$ |
|---|-----------|-------|-------|-----------|
| 0 |           | 1000  |       | (1,0)     |
| 1 | 1000      | 18    | 55    | (0,1)     |
| 2 | 18        | 10    | 1     | (1, -55)  |
| 3 | 10        | 8     | 1     | (-1, 56)  |
| 4 | 8         | 2     | 4     | (2, -111) |
| 5 | 2         | 0     |       | (-5,500)  |

Computation

$$1000 = 55 * 18 + 10$$

$$18 = 1 * 10 + 8$$

$$10 = 1 * 8 + 2$$

$$8 = 4 * 2 + 0$$

Since the last nonzero remainder  $r_4 = 2$  and  $\gcd(1000, 18) = 2$ , no inverse of 6 mod 26 exists.

Check

$$2 * 1000 + (-111) * 18 = 2000 - 1998 = 2$$

6.  $1 \leq n \leq m$

**m = 15**

|        |                                                                            |
|--------|----------------------------------------------------------------------------|
| n = 1  | $15 = 15 \cdot 1 + 0 : \gcd = 1$                                           |
| n = 2  | $15 = 7 * 2 + 1, 2 = 2 * 1 + 0 : \gcd = 1$                                 |
| n = 3  | $15 = 5 * 3 + 0 : \gcd = 3$                                                |
| n = 4  | $15 = 3 * 4 + 3, 4 = 1 * 3 + 1, 3 = 3 * 1 + 0 : \gcd = 1$                  |
| n = 5  | $15 = 3 * 5 + 0 : \gcd = 5$                                                |
| n = 6  | $15 = 2 * 6 + 3, 6 = 2 * 3 + 0 : \gcd = 3$                                 |
| n = 7  | $15 = 2 * 7 + 1, 7 = 7 * 1 + 0 : \gcd = 1$                                 |
| n = 8  | $15 = 1 * 8 + 7, 8 = 1 * 7 + 1, 7 = 7 * 1 + 0 : \gcd = 1$                  |
| n = 9  | $15 = 1 * 9 + 6, 9 = 1 * 6 + 3, 6 = 2 * 3 + 0 : \gcd = 3$                  |
| n = 10 | $15 = 1 * 10 + 5, 10 = 2 * 5 + 0 : \gcd = 5$                               |
| n = 11 | $15 = 1 * 11 + 4, 11 = 2 * 4 + 3, 4 = 1 * 3 + 1, 3 = 3 * 1 + 0 : \gcd = 1$ |
| n = 12 | $15 = 1 * 12 + 3, 12 = 4 * 3 + 0 : \gcd = 3$                               |
| n = 13 | $15 = 1 * 13 + 2, 13 = 6 * 2 + 1, 2 = 2 * 1 + 0 : \gcd = 1$                |
| n = 14 | $15 = 1 * 14 + 1, 14 = 14 * 1 + 0 : \gcd = 1$                              |

Coprime n: {1,2,4,7,8,11,13,14}  $\phi(15) = 8$

**m = 17**

|       |                                            |
|-------|--------------------------------------------|
| n = 1 | $17 = 17 * 1 + 0 : \gcd = 1$               |
| n = 2 | $17 = 8 * 2 + 1, 2 = 2 * 1 + 0 : \gcd = 1$ |

|        |                                                                                 |
|--------|---------------------------------------------------------------------------------|
| n = 3  | $17 = 5 * 3 + 2, 3 = 1 * 2 + 1, 2 = 2 * 1 + 0: \text{gcd} = 1$                  |
| n = 4  | $17 = 4 * 4 + 1, 4 = 4 * 1 + 0: \text{gcd} = 1$                                 |
| n = 5  | $17 = 3 * 5 + 2, 5 = 2 * 2 + 1, 2 = 2 * 1 + 0: \text{gcd} = 1$                  |
| n = 6  | $17 = 2 * 6 + 5, 6 = 1 * 5 + 1, 5 = 5 * 1 + 0: \text{gcd} = 1$                  |
| n = 7  | $17 = 2 * 7 + 3, 7 = 2 * 3 + 1, 3 = 3 * 1 + 0: \text{gcd} = 1$                  |
| n = 8  | $17 = 2 * 8 + 1, 8 = 8 * 1 + 0 = 1: \text{gcd} = 1$                             |
| n = 9  | $17 = 1 * 9 + 8, 9 = 1 * 8 + 1, 8 = 8 * 1 + 0: \text{gcd} = 1$                  |
| n = 10 | $17 = 1 * 10 + 7, 10 = 1 * 7 + 3, 7 = 2 * 3 + 1, 3 = 3 * 1 + 0: \text{gcd} = 1$ |
| n = 11 | $17 = 1 * 11 + 6, 11 = 1 * 6 + 5, 6 = 1 * 5 + 1, 5 = 5 * 1 + 0: \text{gcd} = 1$ |
| n = 12 | $17 = 1 * 12 + 5, 12 = 2 * 5 + 2, 5 = 2 * 2 + 1, 2 = 2 * 1 + 0: \text{gcd} = 1$ |
| n = 13 | $17 = 1 * 13 + 4, 13 = 3 * 4 + 1, 4 = 4 * 1 + 0: \text{gcd} = 1$                |
| n = 14 | $17 = 1 * 14 + 3, 14 = 4 * 3 + 2, 3 = 1 * 2 + 1, 2 = 2 * 1 + 0: \text{gcd} = 1$ |
| n = 15 | $17 = 1 * 15 + 2, 15 = 7 * 2 + 1, 2 = 2 * 1 + 0: \text{gcd} = 1$                |
| n = 16 | $17 = 1 * 16 + 1, 16 = 16 * 1 + 0: \text{gcd} = 1$                              |

Coprime n:  $\{1...16\}$   $\partial(17) = 16$

**m = 23**

|        |                                                                                                |
|--------|------------------------------------------------------------------------------------------------|
| n = 1  | $23 = 23 * 1 + 0: \text{gcd} = 1$                                                              |
| n = 2  | $23 = 11 * 2 + 1, 2 = 2 * 1 + 0: \text{gcd} = 1$                                               |
| n = 3  | $23 = 7 * 3 + 2, 3 = 1 * 2 + 1, 2 = 2 * 1 + 0: \text{gcd} = 1$                                 |
| n = 4  | $23 = 5 * 4 + 3, 4 = 1 * 3 + 1, 3 = 3 * 1 + 0: \text{gcd} = 1$                                 |
| n = 5  | $23 = 4 * 5 + 3, 5 = 1 * 3 + 2, 3 = 1 * 2 + 1, 2 = 2 * 1 + 0: \text{gcd} = 1$                  |
| n = 6  | $23 = 3 * 6 + 5, 6 = 1 * 5 + 1, 5 = 5 * 1 + 0: \text{gcd} = 1$                                 |
| n = 7  | $23 = 3 * 7 + 2, 7 = 3 * 2 + 1, 2 = 2 * 1 + 0: \text{gcd} = 1$                                 |
| n = 8  | $23 = 2 * 8 + 7, 8 = 1 * 7 + 1, 7 = 7 * 1 + 0: \text{gcd} = 1$                                 |
| n = 9  | $23 = 2 * 9 + 5, 9 = 1 * 5 + 4, 5 = 1 * 4 + 1, 4 = 4 * 1 + 0: \text{gcd} = 1$                  |
| n = 10 | $23 = 2 * 10 + 3, 10 = 3 * 3 + 1, 3 = 3 * 1 + 0: \text{gcd} = 1$                               |
| n = 11 | $23 = 2 * 11 + 1, 11 = 11 * 1 + 0: \text{gcd} = 1$                                             |
| n = 12 | $23 = 1 * 12 + 11, 12 = 1 * 11 + 1, 11 = 11 * 1 + 0: \text{gcd} = 1$                           |
| n = 13 | $23 = 1 * 13 + 10, 13 = 1 * 10 + 3, 10 = 3 * 3 + 1, 3 = 3 * 1 + 0: \text{gcd} = 1$             |
| n = 14 | $23 = 1 * 14 + 9, 14 = 1 * 9 + 5, 9 = 1 * 5 + 4, 5 = 1 * 4 + 1, 4 = 4 * 1 + 0: \text{gcd} = 1$ |
| n = 15 | $23 = 1 * 15 + 8, 15 = 1 * 8 + 7, 8 = 1 * 7 + 1, 7 = 7 * 1 + 0: \text{gcd} = 1$                |
| n = 16 | $23 = 1 * 16 + 7, 16 = 2 * 7 + 2, 7 = 3 * 2 + 1, 2 = 2 * 1 + 0: \text{gcd} = 1$                |
| n = 17 | $23 = 1 * 17 + 6, 17 = 2 * 6 + 5, 6 = 1 * 5 + 1, 5 = 5 * 1 + 0: \text{gcd} = 1$                |
| n = 18 | $23 = 1 * 18 + 5, 18 = 3 * 5 + 3, 5 = 1 * 3 + 2, 3 = 1 * 2 + 1, 2 = 2 * 1 + 0: \text{gcd} = 1$ |
| n = 19 | $23 = 1 * 19 + 4, 19 = 4 * 4 + 3, 4 = 1 * 3 + 1, 3 = 3 * 1 + 0: \text{gcd} = 1$                |
| n = 20 | $23 = 1 * 20 + 3, 20 = 6 * 3 + 2, 3 = 1 * 2 + 1, 2 = 2 * 1 + 0: \text{gcd} = 1$                |
| n = 21 | $23 = 1 * 21 + 2, 21 = 10 * 2 + 1, 2 = 2 * 1 + 0: \text{gcd} = 1$                              |
| n = 22 | $23 = 1 * 22 + 1, 22 = 22 * 1 + 0: \text{gcd} = 1$                                             |

Coprime n:  $\{1,...,22\}$   $\partial(23) = 22$

7.

7.1

If  $p$  is prime, then its only divisors are 1 and  $p$ . So, for every integer  $n$  such that  $1 \leq n \leq p - 1$ ,  $\gcd(n, p) = 1$ . Because  $p$  does not divide  $n$ .

So, all  $(p-1)$  numbers below  $p$  are coprime to it, giving this formula:

$$\phi(p) = p - 1$$

This shows that primes are the simplest case because they have no smaller factors and every smaller integer shares no common factor with  $p$  except 1.

7.2

$m = p * q$  where  $p$  and  $q$  are distinct primes

count numbers  $n$  in  $\{1, \dots, pq\}$  that are not divisible by  $p$  or  $q$ .

There are  $q$  multiples of  $p(p, 2p, 3p, \dots, qp)$

There are  $p$  multiples of  $q(q, 2q, 3q, \dots, pq)$

The number  $pq$  is counted in both lists so we subtract 1

Total number not coprime to  $pq = p + q - 1$

So total numbers 1 to  $pq = pq$

Therefore:

$$\phi(pq) = pq - (p + q - 1)$$

$$= pq - p - q + 1$$

$$\phi(m) = (p - 1)(q - 1).$$

8.

$$8.1 \ m = 22$$

$$22 = 2 * 11$$

Where  $p = 2, q = 11$

$$\phi(22) = (2 - 1)(11 - 1) = (1)(10) = 10$$

$$\phi(22) = 10$$

$$8.2 \ m = 35$$

$$35 = 5 * 7$$

Where  $p = 5, q = 7$ :

$$\phi(35) = (5 - 1)(7 - 1) = (4)(6) = 24$$

$$\phi(35) = 24$$

9.

9.1.  $a = 3, n = 7$  (prime)

Using FLT:  $a^{p-2} \equiv 1 \pmod{p} \rightarrow a^{p-2} \equiv a^{-1} \pmod{p}$

$$3^{7-2} = 3^5 \pmod{7}$$

Compute:  $3^2 = 9 \equiv 2, 3^4 \equiv 2^2 = 4, 3^5 \equiv 4 * 3 = 12 \equiv 5$ .

Answer:  $3^{-1} \equiv 5 \pmod{7}$

Check:  $3 * 5 = 15 \equiv 1 \pmod{7}$

9.2.  $a = 6, n = 13$  (prime)

Using FLT:  $6^{13-2} = 6^{11} \pmod{13}$

$$6^2 = 36 \equiv 10; 6^4 \equiv 10^2 = 100 \equiv 9; 6^8 \equiv 9^2 = 81 \equiv 3$$

Then  $6^{11} = 6^8 * 6^2 * 6 \equiv 3 * 10 * 6 \equiv (30 \equiv 4) * 6 \equiv 24 \equiv 11$ .

Answer:  $6^{-1} \equiv 11 \pmod{13}$

Check:  $6 * 11 = 66 \equiv 1 \pmod{13}$

9.3.  $a = 7, n = 14$  (composite)

$\gcd(7, 14) = 7 \neq 1$ , so no inverse exists

Check: Any  $7x$  is a multiple of 7 but 1 isn't:  $7 * 0 \equiv 0, 7 * 1 \equiv 7, 7 * 2 \equiv 0$

$\{0, 7\}$  never 1. No inverse.

10.

$$m = 11$$

$\phi(11) = 10$  (11 is prime, so the units are  $\{1, 2, \dots, 10\}$ )

For all  $a \in \{1, \dots, 10\}$  it must have  $a^{10} \equiv 1 \pmod{11}$

$$a = 2: 2^5 = 32 \equiv -1 \pmod{11} \rightarrow 2^{10} = (2^5)^2 \equiv (-1)^2 \equiv 1$$

$$a = 3: 3^5 = 243 \equiv 243 - 22 * 11 = 243 - 242 = 1 \rightarrow 3^{10} = 1^2 \equiv 1$$

$$a = 10 \equiv -1: (-1)^{10} = 1$$

Thus  $a^{10} \equiv 1$  for every  $a$  with  $\gcd(a, 11) = 1$



The only non-unit in  $Z_{11}$  is  $a \equiv 0$ .  $0^{10} = 0 \not\equiv 1 \pmod{11}$ . So Euler's Theorem does not hold when  $\gcd(a, 11) \neq 1$ .

$$m = 16$$

$$\varphi(16) = 8$$

$$\{1, 3, 5, 7, 9, 11, 13, 15\}$$

Show that  $a^8 \equiv 1 \pmod{16}$  for every odd  $a$ ,  $a^4 \equiv 1 \pmod{16}$ , so  $a^8 \equiv 1$

$$1^4 = 1.$$

$$3^2 = 9, 3^4 = 81 \equiv 1$$

$$5^2 = 25 \equiv 9, 5^4 \equiv 9^2 = 81 \equiv 1$$

$$7^2 = 49 \equiv 1, 7^4 \equiv 1$$

$$9 \equiv -7 \rightarrow 9^2 \equiv (-7)^2 = 49 \equiv 1 \rightarrow 9^4 \equiv 1$$

$$11 \equiv -5 \rightarrow 11^2 \equiv 25 \equiv 9 \rightarrow 11^4 \equiv 9^2 = 81 \equiv 1$$

$$13 \equiv -3 \rightarrow 13^2 \equiv 9 \rightarrow 13^4 \equiv 81 \equiv 1$$

$$15 \equiv -1 \rightarrow 15^4 \equiv 1$$

So all units satisfy  $a^8 \equiv 1 \pmod{16}$  so  $a^4 \equiv 1$

Check:

Even  $a$  so  $\gcd(a, 16) \neq 1$

$$a = 2: 2^8 = 256 \equiv 0 \not\equiv 1 \pmod{16}$$

$$a = 4: 4^4 = 256 \equiv 0 \pmod{16} \rightarrow 4^8 \equiv 0$$

So Euler's theorem does not hold when  $\gcd(a, 16) \neq 1$

11.

If  $\gcd(a, n) = 1$ , then

$$a^{\varphi(n)} \equiv 1 \pmod{n}$$

from this, we get

$$a^{\varphi(n)-1} \equiv a^{-1} \pmod{n}.$$

$$n = 26$$

$$26 = 2 * 13$$

$$\phi(26) = \phi(2) * \phi(13) = (2 - 1)(13 - 1) = 12.$$

So Euler's theorem gives  $a^{12} \equiv 1 \pmod{26}$ , for  $\gcd(a, 26) = 1$

Thus,

$$a^{-1} \equiv a^{11} \pmod{26} \text{ not } a^{-1} \equiv a^{15} \pmod{26}$$

Check:

$$a^{-1} \equiv a^{15} \pmod{26}$$

let  $a = 3$  where  $\gcd(3, 26) = 1$

$$3^{11} \equiv 9 \pmod{26}, 3 * 9 \equiv 1$$

But  $3^{15} \equiv 1 \pmod{26}$ , which is not the inverse

So, the correct relation from Euler's Theorem is  $a^{-1} \equiv a^{11} \pmod{26}$

12.

12.1

Calculate totient function  $\phi(n)$

$$\phi(n) = (p - 1)(q - 1)$$

$p = 43$  and  $q = 19$

$$\phi(n) = (43 - 1)(19 - 1) = (42)(18) = 756$$

$1 < e < \phi(n)$  and  $\gcd(e, \phi(n)) = 1$

$e_1 = 33$ :  $\gcd(33, 756)$

$$756 = 33 * 22 + 30$$

$$33 = 30 * 1 + 3$$

$$30 = 3 * 10 + 0 : \gcd = 3 \neq 1, \text{ so is invalid}$$

$e_2 = 50$ :  $\gcd(50, 756)$

$$756 = 50 * 15 + 6$$

$$50 = 6 * 8 + 2$$

$$6 = 2 * 3 + 0 : \gcd = 2 \neq 1, \text{ so is invalid}$$

Neither 33 or 55 are a valid RSA exponent for  $p$  and  $q$ .

12.2

$e = 5$  to show example.

$d = e^{-1} \bmod (\varphi(n)) = 5^{-1} \bmod (756)$  with EEA

$r_i$  with  $r_i = s_i * 756 + t_i * 5$

| <b>i</b> | <b><math>r_{i-1}</math></b> | <b><math>r_i</math></b> | <b><math>q_i</math></b> | <b><math>s_i/t_i</math></b> |
|----------|-----------------------------|-------------------------|-------------------------|-----------------------------|
| <b>0</b> | 756                         | 5                       |                         | (1,0)                       |
| <b>1</b> | 5                           | 1                       | 151                     | (0,1)                       |
| <b>2</b> | 1                           | 0                       | 5                       | (1,-151)                    |

Computations and check

$$756 = 5 * 151 + 1$$

$$5 = 1 * 5 + 0$$

$$1 = 756 - 151 * 5 = (1) * 756 + (-151) * 5$$

So the coefficient of 5 is -151,

$$d \equiv -151 \bmod 756 \equiv 756 - 151 = 605$$

Check:

$$5 * 605 = 3025, \text{ and } 3025 \bmod 756 = 1$$

Keys:

$$n = pq = 43 * 19 = 817, \varphi(n) = 756$$

$$\text{public key}(n,e) = (817,5)$$

$$\text{private key: } K_{pr} = (p,q,d) = (43,19,605)$$

13. Invariant each iteration (bit b):  $R \leftarrow R^2 \bmod m$ ; if  $b = 1$ , then  $R \leftarrow R * x \bmod m$

13.1  $x=4$ ,  $e = 81$ ,  $m = 103$

| <b>Step</b> | <b>bit b</b> | <b>R after square</b>              | <b>if b = 1 then <math>r * 4 \bmod (103)</math></b> | <b>exponent prefix.<br/>Bin -&gt; dec</b> |
|-------------|--------------|------------------------------------|-----------------------------------------------------|-------------------------------------------|
| <b>1</b>    | 1            | $1^2 \equiv 1$                     | $1 * 4 \equiv 4$                                    | 1 -> 1                                    |
| <b>2</b>    | 0            | $4^2 = 16$                         |                                                     | 10 -> 2                                   |
| <b>3</b>    | 1            | $16^2 = 256 - 2 * 103 = 50$        | $50 * 4 = 200 \equiv 97$                            | 101 -> 5                                  |
| <b>4</b>    | 0            | $97^2 = 9409 - 91 * 103 = 36$      |                                                     | 1010 -> 10                                |
| <b>5</b>    | 0            | $36^2 = 1296 - 12 * 103 = 60$      |                                                     | 10100 -> 20                               |
| <b>6</b>    | 0            | $60^2 = 3600 - 34 * 103 \equiv 98$ |                                                     | 101000 -> 40                              |

|   |   |                               |                            |               |
|---|---|-------------------------------|----------------------------|---------------|
| 7 | 1 | $98^2 = 9604 - 93 * 103 = 25$ | $25 * 4 = 100 \equiv 1000$ | 1010001 -> 81 |
|---|---|-------------------------------|----------------------------|---------------|

$$4^{81} \bmod(103) = 100$$

Binary of e:  $81_{10} = 1010001_2$

13.2  $x = 5$ ,  $e = 199$ ,  $m = 103$

| Step | bit b | R after square                     | if b = 1 then $r * 4 \bmod(103)$ | exponent prefix.<br>Bin -> dec |
|------|-------|------------------------------------|----------------------------------|--------------------------------|
| 1    | 1     | $1^2 \equiv 1$                     | $1 * 5 \equiv 5$                 | 1 -> 1                         |
| 2    | 0     | $5^2 = 25$                         | $25 * 5 = 125 \equiv 22$         | 11 -> 3                        |
| 3    | 1     | $22^2 = 484 - 4 * 103 = 72$        |                                  | 110 -> 6                       |
| 4    | 0     | $72^2 = 5184 - 50 * 103 = 34$      |                                  | 1100 -> 12                     |
| 5    | 0     | $34^2 = 1156 - 11 * 103 = 23$      |                                  | 11000 -> 24                    |
| 6    | 0     | $23^2 = 529 - 5 * 103 = 14$        | $14 * 5 = 70 \equiv 70$          | 110001 -> 49                   |
| 7    | 1     | $70^2 = 4900 - 47 * 103 = 59$      | $59 * 5 = 295 \equiv 89$         | 1100011 -> 99                  |
| 8    | 1     | $89^2 = 7921 - 76 * 103 \equiv 93$ | $93 * 5 = 465 \equiv 53$         | 11000111 -> 199                |

$$5^{199} \bmod(103) = 53$$

Binary of e:  $199_{10} = 11000111_2$

14.

14.1  $p = 4$ ,  $q = 13$ ,  $d = 7$ ,  $x = 9$

RSA so  $p$  and  $q$  must be primes.  $p = 4$  is not a prime, so this is not a valid RSA setup and not recommended to use.

$$n = pq \text{ and } \gcd(4,13) = 1$$

$$n = 4 * 13 = 52$$

$$\phi(52) = \phi(4) \phi(13) = 2 * 12 = 24$$

$$\text{need } ed \equiv 1 \pmod{\phi(n)} \rightarrow e \equiv d^{-1} \pmod{24}$$

EEA to invert  $d = 7 \bmod(24)$

$$24 = 7 * 3 + 3$$

$$7 = 3 * 2 + 1$$

$$3 = 1 * 3 + 0$$

Substitute back in

$$1 = 7 - 2 * 3 = 7 - 2(24 - 3 * 7) = 7 - 2 * 24 + 6 * 7 = 7 * 7 - 2 * 24$$

$$\text{So } 7^{-1}(\text{mod } 24) = 7$$

$$e = 7$$

Encrypt  $y = x^e \text{ mod } n$  with  $x = 9$

$$9^7 \text{ mod } 52$$

$$9^2 = 81 \equiv 81 - 52 = 29$$

$$9^4 \equiv 29^2 = 841 \equiv 841 - 16 * 52 = 841 - 832 = 9$$

$$9^7 = 9^{4+2+1} \equiv (9^4)(9^2)(9) \equiv 9 * 29 * 9$$

$$9 * 29 = 261 \equiv 261 - 5 * 52 = 261 - 260 = 1$$

$$1 * 9 = 9$$

$$\text{So } y = 9$$

$$\text{Decrypt } x' = y^d \text{ mod } n = 9^7 \text{ mod}(52)$$

Same computation as above so  $x' = 9$

$x = 9$  is coprime to  $n = 52$ ,  $\text{gcd}(9, 52) = 1$ , so inversion happens correctly but doesn't follow requirements of both  $p$  and  $q$  being prime numbers.

$$14.2 \text{ } p = 4, q = 13, d = 7, x = 34$$

$$n, \phi(n), e$$

same as 14.1

$$n = 52, \phi(n) = 24, e = 7$$

Encrypt  $y = x^e \text{ mod } n$  with  $x = 34$

$$34^7 \text{ mod } (52)$$

$$34^2 = 1156 \equiv 1156 - 22 * 52 = 1156 - 1144 = 12$$

$$34^4 = 12^2 = 144 \equiv 144 - 2 * 52 = 40$$

$$34^7 = 34^{4+2+1} \equiv 40 * 12 * 34$$

$$40 * 12 = 480 \equiv 480 - 9 * 52 = 480 - 468 = 12$$

$$12 * 34 = 408 \equiv 408 - 7 * 52 = 408 - 364 = 44$$

So  $y = 44$

Decrypt  $x' = y^d \bmod n = 44^7 \bmod 52$

$$44^2 = 1936 \equiv 1936 - 37 * 52 = 1936 - 1924 = 12$$

$$44^4 \equiv 12^2 = 144 \equiv 40$$

$$44^7 \equiv 40 * 12 * 44$$

$$40 * 12 = 480 \equiv 12$$

$$12 * 44 = 528 \equiv 528 - 10 * 52 = 528 - 520 = 8$$

So  $x' = 8 \neq 34$

Decryption failed, since  $\gcd(34, 52) = 2 \neq 1$ .  $x$  is not a unit mod  $n$ . Also both  $p$  and  $q$  are not prime so it is not a valid RSA setup.

14.3  $p = 6, q = 13, e = 5, x = 12$

$n, \phi(n)$

$$n = 6 * 13 = 78$$

$$\phi(78) = \phi(6) \phi(13) = 2 * 12 = 24$$

$$d \equiv e^{-1} \bmod 24 \text{ with } e = 5$$

EEA to invert 5 mod (24)

$$24 = 5 * 4 + 4$$

$$5 = 4 * 1 + 1$$

$$4 = 1 * 4 + 0$$

Substitute it back in

$$1 = 5 - 1 * 4 = 5 - (24 - 4 * 5) = 5 * 5 - 24$$

So,  $5^{-1} \bmod(24) = 5$

$$d = 5$$

encrypt  $y = x^e \bmod n = 12^5 \bmod(78)$

$$12^2 = 144 \equiv 144 - 78 = 66$$

$$12^4 \equiv 66^2 = 4356 \equiv 4356 - 55 * 78 = 4356 - 4290 = 66$$

$$12^5 = 12^4 * 12 \equiv 66 * 12 = 792 \equiv 792 - 10 * 78 = 792 - 780 = 12$$

So  $y = 12$

Decrypt  $x' = y^d \bmod n = 12^5 \bmod 78$

Same calculation so  $x = 12$ .

So  $y = 12$

Although  $p$  is not prime it has inversion happen correctly, but since its not prime it doesn't follow RSA so not a valid RSA system, not recommended to use.